

The closed form for $S_n = \sum_{\ell=0}^{n-1} (\ell^2 \bmod n)$
a step-by-step derivation

1 The quantity and the goal

For an integer $n \geq 2$ we study

$$S_n = \sum_{\ell=0}^{n-1} (\ell^2 \bmod n),$$

where $\ell^2 \bmod n \in \{0, 1, \dots, n-1\}$ is the least nonnegative remainder of ℓ^2 on division by n .

1.1 An equivalent “by value” description

Instead of summing over ℓ , we may sum over the possible *values* $j = \ell^2 \bmod n$. If $N(j) = \#\{x \bmod n : x^2 \equiv j\}$ is the number of square roots of j , then collecting equal values,

$$S_n = \sum_{j=0}^{n-1} j N(j).$$

This viewpoint explains where the arithmetic enters: $N(j)$ is a multiplicative, character-theoretic object, and S_n is its first moment. We will not use this form directly, but it is the reason class numbers appear.

1.2 Notation fixed once and for all

Throughout, set

$$\zeta = e^{2\pi i/n}, \quad n = m^2 d \quad (d \text{ squarefree}), \quad G(t) = \sum_{\ell=0}^{n-1} \zeta^{t\ell^2}.$$

Here m^2 is the largest square dividing n , so $m = \prod_{p^a \parallel n} p^{\lfloor a/2 \rfloor}$; and $G(t)$ is a *quadratic Gauss sum*. We will prove:

Theorem 1.

$$S_n = \frac{n(n-1)}{2} - \frac{n(m-1)}{2} - 2n \sum_{\substack{D < 0 \\ \text{fundamental} \\ D \mid n}} \frac{h_{\mathbb{Q}(\sqrt{D})}}{w_D} \sum_{D \cdot s^2 \mid n} s \prod_{p \mid s} \left(1 - \frac{1}{p} \left(\frac{D}{p}\right)\right),$$

where D runs over negative fundamental discriminants dividing n , $h_{\mathbb{Q}(\sqrt{D})}$ is the class number and $w_D = |\mathcal{O}_{\mathbb{Q}(\sqrt{D})}^\times| \in \{2, 4, 6\}$ the number of units of $\mathbb{Q}(\sqrt{D})$, and $\left(\frac{D}{p}\right)$ is the Kronecker symbol.

2 Strategy

The obstacle is the operation “mod n ,” which is not algebraic. The plan is to *linearise* it:

1. Expand the residue function $a \mapsto (a \bmod n)$ in the additive characters ζ^{ta} of $\mathbb{Z}/n$ (finite Fourier analysis). This converts the awkward “mod” into a clean sum of exponentials.
2. Substituting $a = \ell^2$ and summing over ℓ replaces $\sum_{\ell} \zeta^{t\ell^2}$ by the Gauss sum $G(t)$, a thoroughly understood object.
3. Gauss’s evaluation of $G(t)$ produces a *quadratic character*, and Dirichlet’s class number formula turns the remaining sum into class numbers.

Steps 1–2 are elementary; the two deep classical theorems live in Step 3.

3 The Fourier identity

3.1 Finite Fourier analysis on $\mathbb{Z}/n$

Every function $f : \mathbb{Z}/n \rightarrow \mathbb{C}$ has a unique expansion

$$f(a) = \sum_{t=0}^{n-1} \hat{f}(t) \zeta^{ta}, \quad \hat{f}(t) = \frac{1}{n} \sum_{a=0}^{n-1} f(a) \zeta^{-ta},$$

because the characters $\{\zeta^{ta}\}_t$ are orthogonal: $\sum_{a=0}^{n-1} \zeta^{(t-t')a} = n$ if $t \equiv t'$ and 0 otherwise (a geometric series whose ratio $\zeta^{t-t'}$ is an n -th root of unity $\neq 1$ when $t \not\equiv t'$). We apply this to the specific function $f(a) = a \bmod n$, i.e. $f(a) = a$ for $a \in \{0, \dots, n-1\}$.

3.2 A geometric-sum lemma

We need the coefficients $\hat{f}(t)$, which require $\sum_{a=0}^{n-1} a \omega^a$ for a root of unity ω .

Lemma 1. *If $\omega^n = 1$ and $\omega \neq 1$, then $\sum_{a=0}^{n-1} a \omega^a = \frac{n}{\omega - 1}$.*

Proof. Differentiate the geometric sum $\sum_{a=0}^{n-1} x^a = \frac{x^n - 1}{x - 1}$ and multiply by x :

$$\sum_{a=0}^{n-1} a x^a = \frac{nx^n(x-1) - x(x^n-1)}{(x-1)^2}.$$

Set $x = \omega$ and use $\omega^n = 1$: the numerator becomes $n(\omega - 1) - \omega \cdot 0 = n(\omega - 1)$, so the right side is $n(\omega - 1)/(\omega - 1)^2 = n/(\omega - 1)$. $\square$

3.3 Fourier coefficients of the residue function

The $t = 0$ coefficient is the average value, $\hat{f}(0) = \frac{1}{n} \sum_{a=0}^{n-1} a = \frac{n-1}{2}$. For $t \neq 0$, Lemma 1 with $\omega = \zeta^{-t}$ gives

$$\hat{f}(t) = \frac{1}{n} \sum_{a=0}^{n-1} a \zeta^{-ta} = \frac{1}{n} \cdot \frac{n}{\zeta^{-t} - 1} = \frac{1}{\zeta^{-t} - 1}.$$

Therefore, for every a ,

$$a \bmod n = \frac{n-1}{2} + \sum_{t=1}^{n-1} \frac{\zeta^{ta}}{\zeta^{-t} - 1}. \quad (1)$$

This is the key linearisation: the left side is a “sawtooth,” the right side a finite trigonometric sum.

3.4 Summing over ℓ : enter the Gauss sum

Put $a = \ell^2$ in (1) and sum over $\ell = 0, \dots, n-1$. The constant term contributes $n \cdot \frac{n-1}{2} = \binom{n}{2}$, and interchanging the two finite sums,

$$S_n = \binom{n}{2} + \sum_{t=1}^{n-1} \frac{1}{\zeta^{-t} - 1} \sum_{\ell=0}^{n-1} \zeta^{t\ell^2} = \binom{n}{2} + \sum_{t=1}^{n-1} \frac{G(t)}{\zeta^{-t} - 1}. \quad (2)$$

The inner sum is precisely the Gauss sum $G(t)$. Everything arithmetic is now hidden in $G(t)$.

3.5 The cotangent identity

We simplify the prefactor $\frac{1}{\zeta^{-t} - 1}$. Writing $\theta = \pi t/n$ so that $\zeta^{-t} = e^{-2i\theta}$, and multiplying numerator and denominator by $e^{i\theta}$,

$$\frac{1}{e^{-2i\theta} - 1} = \frac{e^{i\theta}}{e^{-i\theta} - e^{i\theta}} = \frac{\cos \theta + i \sin \theta}{-2i \sin \theta} = -\frac{1}{2} + \frac{i}{2} \cot \theta.$$

Hence

$$\frac{1}{\zeta^{-t} - 1} = -\frac{1}{2} + \frac{i}{2} \cot \frac{\pi t}{n}. \quad (3)$$

The real constant $-\frac{1}{2}$ will produce an elementary term; the imaginary cot part will produce the class-number term.

3.6 Separating the two contributions

Insert (3) into (2):

$$S_n - \binom{n}{2} = -\frac{1}{2} \sum_{t=1}^{n-1} G(t) + \frac{i}{2} \sum_{t=1}^{n-1} G(t) \cot \frac{\pi t}{n}. \quad (4)$$

We treat the two sums separately.

The constant sum

Summing $G(t)$ over *all* t and swapping order,

$$\sum_{t=0}^{n-1} G(t) = \sum_{\ell=0}^{n-1} \sum_{t=0}^{n-1} \zeta^{t\ell^2} = \sum_{\ell=0}^{n-1} n [n \mid \ell^2] = n N_0, \quad N_0 := \#\{x \bmod n : x^2 \equiv 0\},$$

using $\sum_{t=0}^{n-1} \zeta^{tk} = n$ when $n \mid k$ and 0 otherwise. Since $G(0) = n$, we get $\sum_{t=1}^{n-1} G(t) = nN_0 - n$, so the first sum in (4) equals

$$A := -\frac{1}{2} \sum_{t=1}^{n-1} G(t) = \frac{n(1 - N_0)}{2}. \quad (5)$$

The cotangent sum: only the imaginary part survives

Two symmetries collapse the second sum. First, replacing ℓ by anything does not matter, but replacing t by $n - t$ does:

$$G(n - t) = \sum_{\ell} \zeta^{(n-t)\ell^2} = \sum_{\ell} \zeta^{-t\ell^2} = \overline{G(t)}, \quad \cot \frac{\pi(n-t)}{n} = \cot\left(\pi - \frac{\pi t}{n}\right) = -\cot \frac{\pi t}{n}.$$

Now split $G(t) = \operatorname{Re} G(t) + i \operatorname{Im} G(t)$. Under $t \mapsto n - t$, $\operatorname{Re} G$ is *even* and $\operatorname{Im} G$ is *odd*, while $\cot$ is odd. Thus $\operatorname{Re} G(t) \cot \frac{\pi t}{n}$ is odd in t and sums to 0, whereas $\operatorname{Im} G(t) \cot \frac{\pi t}{n}$ is even and survives:

$$\sum_{t=1}^{n-1} G(t) \cot \frac{\pi t}{n} = i \sum_{t=1}^{n-1} \operatorname{Im} G(t) \cot \frac{\pi t}{n}.$$

Multiplying by $\frac{i}{2}$ turns the leading i into $-\frac{1}{2}$, so the second sum in (4) is

$$B := -\frac{1}{2} \sum_{t=1}^{n-1} \operatorname{Im} G(t) \cot \frac{\pi t}{n}. \quad (6)$$

3.7 The master decomposition

Combining (4), (5), (6):

$$\boxed{S_n - \binom{n}{2} = A + B, \quad A = \frac{n(1 - N_0)}{2}, \quad B = -\frac{1}{2} \sum_{t=1}^{n-1} \operatorname{Im} G(t) \cot \frac{\pi t}{n}.} \quad (7)$$

Everything elementary is in A ; everything deep is in B .

4 The term A : the squareful correction

We must count $N_0 = \#\{x \bmod n : x^2 \equiv 0\}$. By the Chinese Remainder Theorem this factors over the prime powers $p^a \parallel n$, and $x^2 \equiv 0 \pmod{p^a}$ exactly when $p^{\lceil a/2 \rceil} \mid x$. The number of such x modulo p^a is $p^a / p^{\lceil a/2 \rceil} = p^{\lfloor a/2 \rfloor}$. Multiplying,

$$N_0 = \prod_{p^a \parallel n} p^{\lfloor a/2 \rfloor} = m.$$

Hence

$$A = \frac{n(1 - m)}{2} = -\frac{n(m - 1)}{2}. \quad (8)$$

In words: the squares put m -fold extra weight on the residue 0, and this is the only purely elementary deviation from the average. It vanishes precisely when n is squarefree ($m = 1$).

5 The term B : the class-number correction

This is the heart of the proof. We evaluate $\sum_{t=1}^{n-1} \operatorname{Im} G(t) \cot \frac{\pi t}{n}$ in five steps.

5.1 Step 1 — reduction to primitive Gauss sums

Let $g = \gcd(t, n)$ and write $n = gM$, $t = gc$ with $\gcd(c, M) = 1$. In $G(t) = \sum_{k=0}^{n-1} e^{2\pi i g c k^2 / (gM)} = \sum_{k=0}^{n-1} e^{2\pi i c k^2 / M}$ the summand depends only on $k \bmod M$, and as k ranges over $0, \dots, gM - 1$ each residue modulo M occurs g times. Therefore

$$G(t) = g g(c; M), \quad g(c; M) := \sum_{k=0}^{M-1} e^{2\pi i c k^2 / M}, \quad \cot \frac{\pi t}{n} = \cot \frac{\pi c}{M}.$$

So every $G(t)$ is a rational multiple of a *primitive* Gauss sum $g(c; M)$, $\gcd(c, M) = 1$, where $M = n / \gcd(t, n)$ is a divisor of n . This is the natural “denominator” of the angle $\pi t / n$.

5.2 Step 2 — Gauss’s sign theorem

The exact value of $g(c; M)$ — including the notoriously subtle *sign* — was determined by Gauss. For $\gcd(c, M) = 1$,

$$g(c; M) = \begin{cases} \sqrt{M} \left(\frac{c}{M} \right) & M \equiv 1 \pmod{4}, \\ i \sqrt{M} \left(\frac{c}{M} \right) & M \equiv 3 \pmod{4}, \\ 0 & M \equiv 2 \pmod{4}, \\ (1 + i^c) \left(\frac{M}{c} \right) \sqrt{M} & M \equiv 0 \pmod{4}. \end{cases}$$

Taking imaginary parts,

$$\operatorname{Im} g(c; M) = \begin{cases} 0 & M \equiv 1, 2 \pmod{4}, \\ \sqrt{M} \left(\frac{c}{M} \right) & M \equiv 3 \pmod{4}, \\ \sqrt{M} \left(\frac{-M}{c} \right) & M \equiv 0 \pmod{4}. \end{cases}$$

The crucial observation: in every nonzero case $\operatorname{Im} g(c; M) = \sqrt{M} \chi_D(c)$, where $\chi_D = \left(\frac{D}{\cdot} \right)$ is an *odd, real, primitive* Dirichlet character — the Kronecker character of an imaginary quadratic field $\mathbb{Q}(\sqrt{D})$, $D < 0$ a fundamental discriminant. (Concretely $D = -M$ when M is squarefree $\equiv 3 \pmod{4}$; for $4 \mid M$ the symbol $\left(\frac{-M}{\cdot} \right)$ reduces to the character of the squarefree-kernel discriminant.) This is where the field $\mathbb{Q}(\sqrt{D})$ is born: it is forced on us by the sign of the Gauss sum.

5.3 Step 3 — which modulus M feeds which discriminant

A character χ_D has *conductor* $|D|$ (its smallest period). Comparing conductors on both sides of $\operatorname{Im} g(c; M) = \sqrt{M} \chi_D(c)$ shows that the modulus M produces χ_D exactly when M is the conductor times a perfect square:

$$M = |D| s^2, \quad s \geq 1.$$

Intuitively: the “squarefull part” s^2 of M is invisible to the primitive character χ_D (it only affects which residues are coprime to M), so it can be any square, while the squarefree skeleton of M is pinned to the conductor $|D|$.

5.4 Step 4 — grouping the sum

We now organise $\sum_t \operatorname{Im} G(t) \cot \frac{\pi t}{n}$ by the modulus $M = n / \gcd(t, n)$, then by D . For a fixed $M \mid n$, the t ’s with $n / \gcd(t, n) = M$ are exactly $t = \frac{n}{M} c$ with $\gcd(c, M) = 1$; for these $G(t) = \frac{n}{M} g(c; M)$

and $\cot \frac{\pi t}{n} = \cot \frac{\pi c}{M}$. Hence the M -block contributes $\frac{n}{M} \sum_{\gcd(c,M)=1} \operatorname{Im} g(c; M) \cot \frac{\pi c}{M}$. Substituting Step 2 and Step 3,

$$\sum_{t=1}^{n-1} \operatorname{Im} G(t) \cot \frac{\pi t}{n} = \sum_{\substack{D < 0 \\ D|n}} \sum_{\substack{\text{fund.} \\ |D|s^2|n}} \frac{n}{|D|s^2} \sqrt{|D|s^2} T_D(s),$$

where we abbreviate the *finite cotangent sum at level $M = |D|s^2$*

$$T_D(s) := \sum_{\substack{c \bmod |D|s^2 \\ \gcd(c, |D|s^2)=1}} \chi_D(c) \cot \frac{\pi c}{|D|s^2}.$$

5.5 Step 5 — Dirichlet's class number formula

Two classical facts evaluate $T_D(s)$.

The value of $L(1, \chi_D)$. Dirichlet's class number formula states that, for the character χ_D of the imaginary quadratic field $\mathbb{Q}(\sqrt{D})$,

$$L(1, \chi_D) = \frac{2\pi h_{\mathbb{Q}(\sqrt{D})}}{w_D \sqrt{|D|}} > 0, \quad L(s, \chi_D) = \sum_{k \geq 1} \frac{\chi_D(k)}{k^s}.$$

Here $h_{\mathbb{Q}(\sqrt{D})}$ is the class number and w_D the number of units. The positivity $L(1, \chi_D) > 0$ is the single genuinely deep input of the whole proof; no elementary proof of it is known.

The finite cotangent sum as an L -value. For an odd character, $L(1, \cdot)$ has a closed finite expression through cotangents. The level- M sum $T_D(s)$ realises $L(1, \psi_M)$ for the imprimitive character ψ_M modulo $M = |D|s^2$ induced from χ_D ; passing from ψ_M to the primitive χ_D removes the Euler factors at the primes of M outside the conductor:

$$T_D(s) = \frac{2M}{\pi} L(1, \psi_M) = \frac{2M}{\pi} L(1, \chi_D) \prod_{\substack{p|M \\ p \nmid |D|}} \left(1 - \frac{\chi_D(p)}{p}\right).$$

Substituting $M = |D|s^2$ and the value of $L(1, \chi_D)$,

$$T_D(s) = \frac{2|D|s^2}{\pi} \cdot \frac{2\pi h_{\mathbb{Q}(\sqrt{D})}}{w_D \sqrt{|D|}} \prod_{\substack{p|s \\ p \nmid |D|}} \left(1 - \frac{\chi_D(p)}{p}\right) = \frac{4\sqrt{|D|} s^2 h_{\mathbb{Q}(\sqrt{D})}}{w_D} \prod_{\substack{p|s \\ p \nmid |D|}} \left(1 - \frac{\chi_D(p)}{p}\right).$$

(The primes $p \mid M$ outside the conductor are exactly the primes $p \mid s$ with $p \nmid |D|$.) Finally, since $\chi_D(p) = \left(\frac{D}{p}\right) = 0$ whenever $p \mid D$, the factor for such p is 1, so the restriction $p \nmid |D|$ is vacuous and the product may run over *all* $p \mid s$.

5.6 Assembly of B

Put $\sqrt{|D|s^2} = \sqrt{|D|}s$ and the formula for $T_D(s)$ into the grouped sum of Step 4. The (D, s) term is

$$\frac{n}{|D|s^2} \cdot \sqrt{|D|}s \cdot \frac{4\sqrt{|D|}s^2 h_{\mathbb{Q}(\sqrt{D})}}{w_D} \prod_{p|s} \left(1 - \frac{1}{p} \left(\frac{D}{p}\right)\right) = \frac{4n h_{\mathbb{Q}(\sqrt{D})}}{w_D} s \prod_{p|s} \left(1 - \frac{1}{p} \left(\frac{D}{p}\right)\right),$$

where the two powers of $\sqrt{|D|}$ and of s cancel against the denominator $|D|s^2$. Therefore

$$\sum_{t=1}^{n-1} \operatorname{Im} G(t) \cot \frac{\pi t}{n} = \sum_{\substack{D < 0 \text{ fund.} \\ D|n}} \frac{4n h_{\mathbb{Q}(\sqrt{D})}}{w_D} \sum_{D \cdot s^2 | n} s \prod_{p|s} \left(1 - \frac{1}{p} \left(\frac{D}{p}\right)\right),$$

and, using $B = -\frac{1}{2}(\dots)$ from (6),

$$B = -2n \sum_{\substack{D < 0 \text{ fund.} \\ D|n}} \frac{h_{\mathbb{Q}(\sqrt{D})}}{w_D} \sum_{D \cdot s^2 | n} s \prod_{p|s} \left(1 - \frac{1}{p} \left(\frac{D}{p}\right)\right). \quad (9)$$

6 Conclusion

Inserting A from (8) and B from (9) into the master decomposition (7) gives

$$S_n = \binom{n}{2} - \frac{n(m-1)}{2} - 2n \sum_{\substack{D < 0 \text{ fund.} \\ D|n}} \frac{h_{\mathbb{Q}(\sqrt{D})}}{w_D} \sum_{D \cdot s^2 | n} s \prod_{p|s} \left(1 - \frac{1}{p} \left(\frac{D}{p}\right)\right),$$

which is Theorem 1 (with $\binom{n}{2} = \frac{n(n-1)}{2}$). □

Two remarks

An inequality for free. Every class number satisfies $h_{\mathbb{Q}(\sqrt{D})} \geq 1$, every Euler factor $1 - \frac{1}{p} \left(\frac{D}{p}\right) \in \{1 - \frac{1}{p}, 1, 1 + \frac{1}{p}\}$ is positive, and $m \geq 1$; hence both corrections are ≥ 0 and $S_n \leq \binom{n}{2}$, with equality iff n is squarefree and has no divisor $\equiv 3 \pmod{4}$.

Where the depth sits. Steps 1–4 of B , and all of A , are elementary. Only two classical theorems are used as black boxes: Gauss's sign evaluation of $g(c; M)$ (§5.2) and Dirichlet's class number formula (§5.5). The lone non-elementary fact inside them is $L(1, \chi_D) > 0$.